II. (15 points) Computer simulations display astronomical or microscopic motion by scaling geometrical dimensions. For the displayed motion to be physically possible in the scaled system, time must also be scaled.

1. A Newtonian particle orbits in a force field $\boldsymbol{F}(\boldsymbol{r})$ satisfying $\boldsymbol{F}(\alpha\boldsymbol{r}) = \alpha^k \boldsymbol{F}(\boldsymbol{r})$,

k is constant; $\boldsymbol{r}$ is the position vector. For dimension scale α, find the time scale β.

2. Hence show that a planet's orbital period squared scales as its orbit's size cubed.